CHIMDI WALTER NDUBUISI

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RESEARCH INTERESTS

Computer vision and machine learning for scientific imaging; geometry- and topology-aware representation learning; physics-informed segmentation and active contour methods; high-fidelity lesion segmentation for plant phenotyping; biological image analysis; self-supervised and transfer learning for foundation models; diffusion models for structured vision tasks; multimodal learning for biomolecular complexes and protein structure; scalable training and deployment on HPC systems; information geometry; neuro-symbolic reasoning and DSL program synthesis.

EDUCATION

Ph.D., Electrical & Computer Engineering, University of Missouri, Columbia, MO — **Jul 2021 – Present (Expected July 2028)**

Dissertation: Topology-Aware Geometric Deep Learning and Physics-Informed Optimization for Scientific Imaging

• Advisor: Dr. Toni Kazic

Graduate Coursework, Electronics Engineering, University of Maiduguri, Nigeria — **Jan 2020 – Jun 2021**

Completed Master's-level foundational coursework (digital signal processing, microelectronics, communication systems); left the program to begin a direct-entry Ph.D. at the University of Missouri. Degree not conferred.

B.Eng., Electrical & Electronics Engineering, Federal University of Technology, Owerri, Nigeria — **Graduated Dec 2016**

Honors: Second Class Upper • Thesis: Development of a Digital Library Based on a Cloud Computing Model

PROFESSIONAL & RESEARCH EXPERIENCE

Research Affiliate / Intern, Lawrence Berkeley National Laboratory (LBNL) — **Mar 2026 – Present**

Computational Biosciences Group • Supervisor: Dr. Kristofer Bouchard • Berkeley, CA

- I am working toward multimodal protein AI frameworks that combine SAXS profiles, sequence information, predicted structures, and conformer ensembles to learn protein conformational landscapes and characterize biologically relevant structural dynamics.
- Contribute to AI methods for biomolecular complexes in alignment with DOE Genesis-oriented efforts; produce preliminary data supporting prospective grant proposals.
- Building and evaluating a protein SE(3) multimodal large language model for structure-aware reasoning over proteins using sequence, 3D geometry, and biological annotations.
- Initiated TopoFiberFold, a geometry- and topology-aware AlphaFold-style modeling project that integrates protein language model priors, manifold-based geometric learning, and topological scoring for biomolecular complex prediction.
- Exploring differential-geometric and topological techniques, including SE(3)/SO(3) representations, torsional manifolds, and persistent homology for reranking, refinement, and improved plausibility of predicted protein complexes.
- Developing multimodal protein AI frameworks that integrate SAXS profiles, protein sequences, predicted structures, biological annotations, and conformer ensembles to learn conformational landscapes and characterize biologically relevant structural dynamics.
- Building and evaluating a protein-language-model–based multimodal foundation model that integrates amino-acid sequence, protein function, predicted 3D structure, biological annotations, SAXS profiles, and conformational ensembles. The framework uses modality-appropriate tokenization for discrete or symbolic inputs (such as sequence, function, and annotations) while encoding continuous structural and SAXS information through geometry-aware and signal-aware representation modules for joint structure–function reasoning.
- Designing a scalable conformational-ensemble generation pipeline using Cα/Cβ elastic-network and normal-mode displacement methods, complementary sampling strategies, and geometry- and physics-informed validation gates to identify structurally plausible conformers.
- Implementing and benchmarking BioEmu generative deep-learning–based conformational sampling against ProDy ANM/NMA on matched AlphaFold proteins, using locally generated and cached MSAs to compare conformational diversity, structural quality, sampling yield, and runtime; extending the framework toward hybrid BioEmu-guided NMA sampling, where diverse generative conformational states are locally explored through low-frequency normal modes.
- Conducting large-scale ablation and calibration studies across network cutoff, mode selection, displacement amplitude, RMSD constraints, sampling strategy, energy weighting, and ensemble size to derive protein-specific production settings

for eventual scaling from hundreds to millions of proteins.
- Generating synthetic per-conformer and ensemble SAXS data using Cα/full-atom Debye calculations and CRYSOL, and validating computed curves against experimental SAXS data where available.
- Contributing to AI methods for biomolecular complexes in alignment with DOE Genesis-oriented efforts and producing preliminary results to support prospective grant proposals.
- Building and evaluating an SE(3)-aware multimodal protein foundation model for structure-aware reasoning over protein sequence, 3D geometry, SAXS, conformational ensembles, and biological annotations.
- Initiated TopoFiberFold, a geometry- and topology-aware AlphaFold-style modeling project integrating protein language-model priors, manifold-based geometric learning, and topological scoring for biomolecular-complex prediction.
- Exploring differential-geometric and topological techniques, including SE(3)/SO(3) representations, torsional manifolds, and persistent homology, for conformer reranking, structural refinement, and improved plausibility of predicted protein complexes.

Lead Innovator, AI & Scientific R&D, NurseryCorp **Aug 2026 – Present**

Equity Stakeholder • Remote

- Lead AI and scientific R&D initiatives, translating advances in computer vision, multimodal AI, representation learning, scientific machine learning, and computational modeling into commercially relevant technologies.
- Conceive and develop novel, potentially patentable technology from research hypothesis through prototype, technical validation, intellectual-property development, and productization.
- Shape the company's AI innovation strategy and technical roadmap, identifying opportunities for advanced AI across NurseryCorp's scientific, agricultural, health, manufacturing, and digital-technology initiatives.

Research Fellow & Assistant, University of Missouri **Aug 2021 – Present**

Department of Electrical Engineering & Computer Science • Columbia, MO

- Designed product-manifold latent spaces (hyperbolic + Euclidean + spherical) with lightweight decoders improving Dice while preserving global topology; ablated across ResNet-50, DeepLabv3, ViT/DINOv2-v3.
- Built containerized CV pipeline integrating wavelets, Navier-Stokes-inspired diffusion, active contours, SSL/SAM; packaged as Docker-to-Apptainer for reproducible SLURM jobs on Hellbender, Lewis & Clark, and Nautilus.
- Implemented 16-bit color-checker calibration via linear-sum assignment (Hungarian Algorithm) to stabilize segmentation across sessions/devices; distilled a 7M-parameter U-Net for <15 ms single-GPU inference (Dice + TopoLoss).
- Developed early VLM-guided self-distillation: text cues from image+mask pairs looped back into training to improve lesion semantics.
- Lesion-to-biochemical network inference: fused morphological, colorimetric, and topological descriptors (Betti numbers, persistence diagrams) with environmental/genetic covariates to learn sparse physiological interaction graphs; explored sparse DAGs/PGMs and differentiable simulators with counterfactual checks.
- Orchestrated HPC experiments with ablations over geometry/topology weights, backbones, and layer-unfreezing strategies; wrote CLI wrappers, unit tests, and reproducible SLURM job scripts.
- Implemented GPU-accelerated Navier+Snake optimization framework for lesion segmentation (PyTorch + OpenCV + scikit-image), with a robust objective balancing target lesion area fraction, compactness, and suppression of spurious components.
- Trained a CNN regressor to predict segmentation parameters directly from images, reducing expensive per-image search to a single forward pass with light local refinement.
- Agentic LLMs & neuro-symbolic reasoning: built an ARC-AGI-2 hybrid solver (concise DSL for grid transformations, symbolic-first planner, LLM fallback, ViT perception stack; ~65% pixel accuracy on unseen test data) and ReAct-style agentic tool use with structured planning and PPO/DPO-style preference tuning.
- Applied NLP (eBay ML Challenge, 2025): trained XLM-RoBERTa token-classification models with label smoothing, warmup, and span post-processing, achieving ~0.85+ F1 on held-out validation and ranking above the official baseline.

ICT Staff, University of Maiduguri ICT / Computer Centre **Nov 2017 – Dec 2019**

Maiduguri, Nigeria

- Developed an improved database management system and a new university website.
- Managed the university's computer networks and database systems, supporting campus-wide IT infrastructure and operations.

Engineering Trainee, Sunnex Solar **Dec 2016 – Nov 2017**

Nigeria

- Trained in the design of solar and energy systems, including system sizing, component selection, and installation fundamentals.

TEACHING EXPERIENCE

Teaching Assistant, University of Missouri **Aug 2022 – Present**

- CS 1050, Algorithm Design & Programming (C): labs, tutoring, grading, and algorithmic problem-solving.
- CS 3530, Unix Systems: command-line tooling, systems workflows, and project support.

PUBLICATIONS & PREPRINTS

Peer-Reviewed

- Ndubuisi, C.W., Kazic, T., Kharismawati, D.E. (2023). Imaging Maize Lesions. *North American Plant Phenotyping Network (NAPPN).* DOI: 10.22541/au.166505708.85410753/v1
- Somoye, O.I., Ndubuisi, C.W., Donatus, S., Amakye, F. (2025). Scalable Machine Learning Algorithms for Processing High-Dimensional Data: A Case Study Approach. *Asian Journal of Advanced Research and Reports.* DOI: 10.9734/ajarr/2025/v19i61064
- Eleweke, I., Umakor, M.F., Ndubuisi, C.W., et al. (2025). AI-Driven Threat Detection and Prevention in Cloud Computing Environments. *AJISE.*
- Ndubuisi, C.W. and Kazic, T. (2026). HyperTopo-Adapters: Geometry- and Topology-Aware Segmentation of Leaf Lesions on Frozen Encoders. arXiv:2601.06067.

Preprints & Manuscripts in Preparation

- Ndubuisi, C.W. (2026). Geometric Decoding of Pathology Foundation Models Reveals Morphology-Encoded Gene Programmes in Breast Cancer. [Link]
- Ndubuisi, C.W. (2026). Task-Sufficient Manifold Distillation of Learned Representations. [Link]
- Ndubuisi, C.W. (2026). When Does Nonlinearity Matter? Structural Accessibility and the Geometry–Dynamics–Interpretability Tradeoff in Single-Cell Modeling. [Link]
- Ndubuisi, C.W. (2026). Ontology-Guided Pathway Activity Identifies a Cell-Intrinsic Defense Response Program Associated with MEK Inhibitor Sensitivity. *bioRxiv.* DOI: 10.64898/2026.07.25.740732
- Ndubuisi, C.W. (2026). Dataset-Dependent Utility of Discrete Ricci Curvature for Transition-Region Ranking in Single-Cell Lineage Graphs. *bioRxiv.* DOI: 10.64898/2026.07.25.740737
- Ndubuisi, C.W. (2026). Geometric-Relational Diagnosis Spaces for Multi-LLM Clinical Reasoning. *Submitted to NeurIPS 2026; under rebuttal.* [Link]
- Ndubuisi, C.W. (2026). Manifold-Guarded Ranking for Auditable Biomedical Discovery Agents. *Submitted to NeurIPS 2026; under rebuttal.* [Link]
- Ndubuisi, C.W. (2026). Annotation Quality Dominates Architecture Choice in Maize Lesion Segmentation: A 65-Run Benchmark with Domain-Specialised DINOv2 Pre-Training and Dual Human-Corrected Evaluation. [Link]
- Ndubuisi, C.W. (2026). A Reproducible Benchmark of Geometry-Aware Causal Discovery for Genome-Scale Perturb-seq. [Link]
- Ndubuisi, C.W. and Olaoye, Dotun. (2026). Geometry-Sensitive Persistent Representatives for Omics: A Sparse Extension of Harmonic Persistent Homology with Multi-Backend Geometry. [Link]
- Ndubuisi, C.W. and Kazic, T. Gradient-Free Stochastic Optimization of Navier-Stokes Active Contours for Unsupervised High-Throughput Phenotyping. [Link]
- Ndubuisi, C.W. and Kazic, T. Selective Semi-Amortized Inference for Physics-Based Lesion Phenotyping: Uncertainty-Guided Runtime-Quality Tradeoffs. [Link]

SELECTED PROJECTS

- **Concerted Abduction (Multi-LLM Differential Diagnosis):** Built a pipeline converting free-text diagnostic outputs from multiple LLMs (OpenAI, Anthropic, Qwen, DeepSeek, Llama) into structured hypotheses, belief masses, and clinical relation graphs; implemented Dempster–Shafer evidence fusion with composite-aware ranking and open-world/credal robustness extensions, plus blinded expert-annotation workflows.
- **Dronezaic (UAV Mosaics & 3D to Edge Robotics):** Mosaics-to-3D fusion with VGGT; ROS 2 messaging to a ground robot; real-time tiling and lightweight segmentation for field tasks. Led the vision/imagery stack and co-developed task schemas and safety interlocks. Farm Robotics Challenge (2026).
- **Unsupervised Lesion Segmentation at Scale:** DINO-v2 embeddings to auto-masks to U-Net distillation with Dice + TopoLoss (50k unlabeled 128×128 patches); single-GPU real-time inference.
- **Geometry-Aware Topological Omics & Spatial Transcriptomics:** Built manuscript-linked pipelines (CC-SPR, CausalManifold, TopoSpatial) for interpretable omics feature selection, Perturb-seq causal discovery, and spatial transcriptomics using persistent homology, sparse cycle representatives, and diffusion/Ricci/DTM geometry with leakage-aware evaluation.
- **HPC Containerized Segmentation Suite:** Consolidated wavelet, Navier-Stokes diffusion, active contours, and SSL/SAM into a monorepo with Docker/Apptainer images and reproducible SLURM jobs on MU Hellbender.
- **QIC AI + Quantum Explorations (IBM Quantum Innovation Center):** Evaluated IBM Quantum cloud resources for accelerating AI/ML workloads in plant phenotyping and optimization.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, MATLAB
- **LLM / Agents:** Transformers, vLLM, ReAct/ToT, program-of-thought & DSL design, PPO/DPO/RLHF
- **Vision / ML:** PyTorch, TensorFlow, DINOv2/v3, SAM/SAM2, DeepLabv3, U-Net, diffusion models, multi-view geometry (COLMAP, VGGT/MVS), OpenCV, scikit-learn
- **Systems & Data:** Docker/Apptainer, SLURM, Git, Spark; experiment tracking & evaluation harnesses
- **Cloud & Compute:** MU Hellbender & Lewis-and-Clark, Nautilus; 4×A100 / 8×V100 nodes; AWS, Azure, Google Cloud

HONORS, LEADERSHIP & SERVICE

- Recipient, IEEE Computer Society WEIGD Student Support Fund, one of 502 applicants selected (2026)
- Selected Mentee, Bayer B4U Mentoring Program, University of Missouri and Bayer (Mar 2026)
- Honorable Mention Award, Nexus Informatics Conference, for an outstanding informatics abstract, Kansas City, MO (Apr 2026)
- Accepted Student, Wolfram Summer Research Institute (Applications of Computation track), Waltham, MA (Jun 28 – Jul 18, 2026)
- Vice-Programming Director, Graduate Professional Council (Aug 2024 – May 2025)
- Finalist (Top 6), ARC-AGI-2 Hackathon, University of Missouri (Nov 2025)
- Second-Best Poster, DAREC Symposium (Mar 2025)
- EECS Departmental Fellowship (Apr 2021)
- Gold Medal & Best African Student, International Junior Science Olympiad (Dec 2010)

TALKS & WORKSHOPS

- Poster: CC-SPR: Sparse Persistent Representatives with Geometry-Aware Backends for Interpretable Omics Feature Selection. ISMB, Washington, DC (Jul 13, 2026)
- Imaging Maize Lesions. NAPPN (Danforth Center) & MU Plant Research Symposium / Corteva (2023)
- MLCAS Workshop on ML for Climate & Agriculture (Japan 2021; Nebraska 2022)

MEMBERSHIP

- IEEE Student Member (St. Louis Section)